

SwissCivicQA-4L

Does Apertus know the Swiss Federal Constitution equally well in all four national languages?

Hack Apertus Online 2026 · Track 1B Swiss Voices · Core Task Intelligence · Model: Apertus 1.5 8B

45.1%

of 488 items answered correctly

23.8%

of facts correct in all four languages

53.2%

of attempted answers are wrong

0.81

AUROC of cross-lingual agreement as hallucination detector

Summary. Citizens ask about their constitutional rights in their own national language, and a “sovereign, multilingual” Swiss model should give the same correct answer in each. SwissCivicQA-4L is a closed-book factual QA set of 122 facts from the Federal Constitution, each asked in German, French, Italian and Romansh (488 items, incl. false-premise traps), with every gold answer verbatim-grounded in the official text of the same language. Apertus 1.5 8B answers 45.1% of all items correctly, ranging from 43.4% to 46.7% across languages, and only 23.8% of facts are answered correctly in all four languages.

Use of Apertus. Model: `swiss-ai/Apertus-v1.5-8B` (text-only GGUF Q8_0 conversion `andreamartin/apertus-v1.5-8b-text-Q8_0-GGUF`). Used for: evaluation. Runs on: local weights, llama.cpp CPU server (16 cores, no GPU).

1. Use case

The Federal Constitution (SR 101) defines how Swiss direct democracy, federalism and fundamental rights work: how many signatures an initiative needs, who elects the Federal Council, which cantons have half a cantonal vote. These facts matter to every voter, are asked in four official versions of equal legal force (DE/FR/IT; RM for the Romansh-speaking population, Art. 70 para. 1), and change rarely, so a model can be expected to know them. A model that is right in German but wrong in Italian or Romansh fails exactly the users a **Swiss** model is meant to serve. The dataset therefore evaluates two things: **core task intelligence** (factuality and hallucination, SimpleQA-style) and **cross-lingual consistency** of that knowledge, which existing benchmarks do not test for Swiss civic knowledge and in particular not in Romansh.

2. Dataset

2.1. Specification

Item	Specification
Unit	One fact asked in 4 languages (DE/FR/IT/RM, Rumantsch Grischun); single-turn, closed-book question
Size	122 facts × 4 = 488 items; 20 facts are false-premise questions (premise contradicts the Constitution)
Answer types	number, entity, list, yes/no, false-premise (see Table 2)
Ground truth	short gold answer + accepted variants per language, citation (article/paragraph), verbatim evidence span from the official text of that language
Source	Fedlex consolidated version 2024-03-03 (in force on 2026-10-16) in four languages, SHA-256 pinned; Art. 127 excluded (amended with effect from 2029)
Files	<code>eval.jsonl</code> (test cases) · <code>responses_*.jsonl</code> (model outputs) · <code>metadata.jsonl</code> (provenance, licence per item)

2.2. Construction and quality control

Facts were selected to cover all main areas of the Constitution (popular rights, Federal Assembly, Federal Council, courts, fundamental rights, languages, federalism, taxes, social security, transport) and to have one unambiguous answer stated in the text. Questions and translations were drafted with AI assistance and adapted to the official

terminology of each language version (e.g. Romansh *Sursilvania/Sutsilvania* for Obwalden/Nidwalden, *dumonda decisiva* for the tie-break question). Quality is enforced mechanically, not by trust:

- ▶ **Evidence check (every item):** the validator parses the four official HTML files down to article/paragraph/letter and requires the item’s evidence span (≥ 8 characters, containing the answer) to occur verbatim in the cited paragraph of the same language. The build fails on any miss; all 488 items pass.
- ▶ **Stability check:** only provisions unchanged between the version in force and the already adopted 2029 consolidation are used.
- ▶ **False-premise design:** each trap negates a fact that is itself in the dataset, so a model that “knows” the fact but accepts the premise is exposed.
- ▶ **No personal data:** items contain only legal facts; there are no data subjects.

2.3. Licensing and provenance

Official enactments are not protected by copyright (Art. 5 para. 1 lit. a Swiss Copyright Act), so the gold answers can be redistributed freely; questions, answer formulations and labels are released under CDLA-Permissive-2.0. Every row of `metadata.jsonl` carries source URL, version, article, paragraph, evidence span, licence and an authoring note. The only human-generated data are the audit labels (Section 3.3); the annotator consent form is in `docs/consent_form_annotator.md`.

3. Evaluation

3.1. Setup

Inference: llama.cpp server (image pinned by digest), temperature 0, seed 42, max 256 tokens, default chat template of the GGUF (`--jinja`); no reasoning traces occur in any response. The prompt is one short instruction in the question’s language (“answer briefly and precisely”) followed by the question; no system prompt, no retrieval.

3.2. Grading

Each response is labelled CORRECT / INCORRECT / NOT_ATTEMPTED with the SimpleQA grader template (adapted: language-aware, list and false-premise rules) by an open-weights judge, `gemma-3-12b-it Q4_K_M`, run locally (temperature 0). A different model family than the one under test is used to avoid self-preference. Metrics follow SimpleQA: accuracy, correct-given-attempted, hallucination rate = incorrect / attempted, F-score; 95% confidence intervals are percentile bootstraps over facts (2000 resamples, seed 42).

3.3. Grading audit

Two independent checks. (1) **Rules:** numbers and yes/no answers are graded deterministically (number normalisation incl. number words in all four languages); the judge agrees with the rules on 138 items with 97.8% agreement (Cohen’s $\kappa = 0.9542$). (2) **Humans:** a stratified random 20% sample (96 items, 24 per language) was labelled **blind**, i.e. without seeing the judge label, by two adult volunteer annotators (pseudonymous IDs A1, A2), using the same written rubric with the official text and the parallel German item shown side by side. Human–human agreement is 88.5% ($\kappa = 0.7833$); the judge agrees with the human consensus¹ in 86.5% of items ($\kappa = 0.7392$), i.e. close to human level. Per language: DE 95.8%, FR 66.7%, IT 95.8%, RM 87.5%.

Judge bias. The judge is slightly lenient: on the sample it rates 47.9% of answers correct, the humans 44.8%. In 7 of the 9 items where both annotators disagree with the judge, the judge accepted an answer that adds a wrong claim, e.g. “a two-thirds majority of both chambers” for declaring a law urgent (Art. 165: majority of the members), or an answer to a false-premise question that does not correct the premise. Five of these nine are French, which explains the lower French agreement. A human-calibrated estimate of overall accuracy is therefore 41.9% instead of 45.1% all conclusions below hold under both.

¹Majority label; on the 11 items where the two annotators disagree, the label of A1 is used.

4. Results

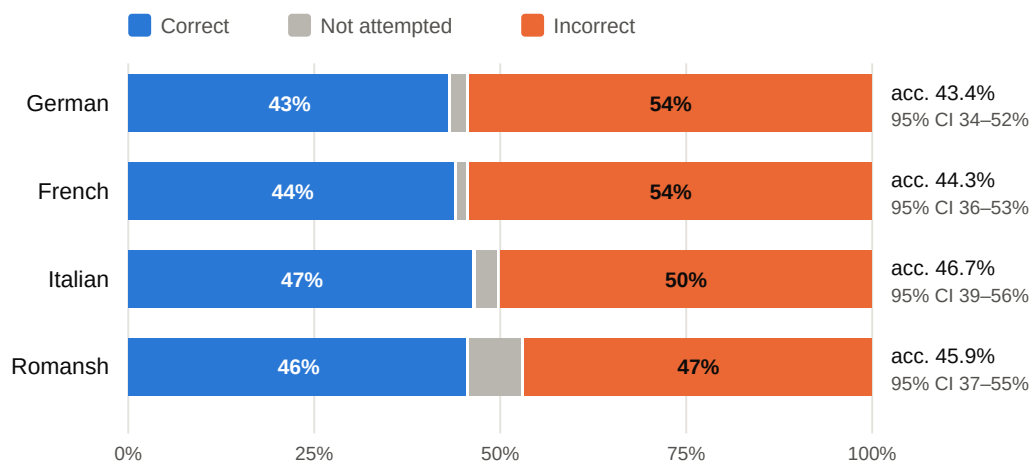


Figure 1: Outcome distribution per language (122 items each). Bars sum to 100%.

Language	Accuracy	95% CI	Correct attempted	Hallucination	Not attempted
German	43.4%	34.4%–52.5%	44.5%	55.5%	3
French	44.3%	36.1%–53.3%	45%	55%	2
Italian	46.7%	38.5%–55.7%	48.3%	51.7%	4
Romansh	45.9%	36.9%–54.9%	49.6%	50.4%	9

Table 1: Main results, Apertus 1.5 8B (Q8_0).

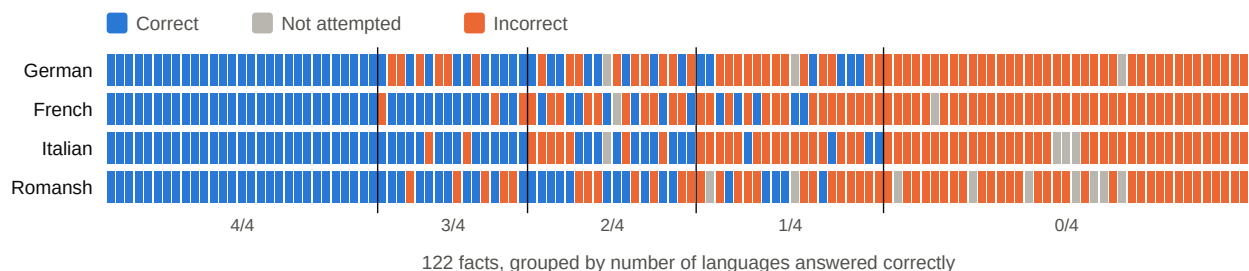


Figure 2: Every fact (column) in every language (row). Only the left block is correct in all four languages: consistency@4 = 23.8%, while 68% of facts are answered correctly in at least one language.

Type	DE	FR	IT	RM
entity	35.9%	47.2%	47.2%	50.9%
false premise	10%	10%	30%	10%
list	42.9%	28.6%	28.6%	14.3%
number	55.2%	55.2%	48.3%	51.7%
yes no	100%	69.2%	76.9%	84.6%

Table 2: Accuracy by answer type and language.

4.1. Findings

1 Equal averages hide unequal knowledge

Accuracy per language lies within a few points (Table 1, overlapping confidence intervals), so an aggregate benchmark would report Apertus as “equally good” in all four languages. At the level of individual facts this is false: 68% of facts are answered correctly in **at least one** language, but only 23.8% in **all four**; 54 facts are known somewhere but not everywhere (Fig. 2). Pairwise label agreement between languages is only 66.4%–76.2%. Example: the 100 000 signatures for an initiative on total revision (Art. 138) are known in German, while French and Italian answer “80 000” and Romansh “100 000 votantas e 60 000 votants”.

2 The model almost never abstains

Only 18 of 488 answers are “not attempted”; when Apertus answers, it is wrong in 53.2% of cases. For a civic-information use case, calibrated abstention would be worth more than a confident wrong number.

3 False premises are accepted, and the inventions differ by language

On false-premise items, accuracy is 10% (DE), 10% (FR), 30% (IT) and 10% (RM). Asked which crimes the Constitution punishes by death (Art. 10 prohibits the death penalty), Apertus answers correctly in German and Italian but in French lists “trahison, haute trahison et meurtre” and in Romansh “tradiment, espionadi ...”. A non-existent “Swiss constitutional court” is said to have annulled federal laws “à 13 reprises” (FR) and “13 leschas federalas” (RM), although Art. 190 makes federal statutes binding on the Federal Supreme Court. English is claimed to be an official language “since 2000” (DE, FR) or “since 1996” (IT, RM).

4 Romansh is not the weakest language on average, but the least predictable

Romansh accuracy is in line with the other languages, and Romansh shows the most abstentions (9). However, facts that are correct in three languages most often fail in German or Romansh, so Romansh users cannot rely on knowledge demonstrated in another language.

5 Simple facts are robust, procedural detail is not

Composition numbers (200 National Council members, 46 Council of States members, 7 Federal Councillors, 4-year terms, 100-day referendum period, 1 August) are correct in all four languages. Errors concentrate on thresholds and qualifiers: the eight cantons that can call a referendum, the six cantons with half a cantonal vote, the 8.5% ceiling on corporate tax and the cantons’ 17% share are wrong in every language, while the SNB’s two-thirds profit share is answered correctly **only** in Romansh. List answers are the weakest answer type (Table 2).

6 Cross-lingual disagreement is a free hallucination detector

Because every fact is asked in four languages, the answers can be compared with each other without knowing the gold answer. For the 140 number and yes/no items, where answers can be compared deterministically, an answer that agrees with the answers in **all** other languages is correct in 89% of cases ($n = 73$), one that agrees with **none** in only 21.4% ($n = 28$). As a correctness predictor, cross-lingual agreement reaches AUROC 0.81. Extending the comparison to all 488 items (an LLM decides whether two answers in different languages state the same fact, without seeing the gold answer), the signal holds: AUROC 0.73 overall (yes/no 0.89, numbers 0.74, entities 0.73), with 68.2% vs. 15.7% correct for full vs. no agreement. It fails on false premises (AUROC 0.46): a premise accepted in every language is consistently wrong, so agreement cannot detect it. For a multilingual deployment this is a cheap, reference-free safeguard: ask in a second national language and flag disagreement.

4.2. Baseline: is the Swiss model better at Swiss facts?

To put the numbers in context we ran the identical pipeline (same items, prompts, decoding, judge) on **Qwen3 8B** (Q8_0, thinking disabled), an open model of the same size that was not trained with a Swiss focus.

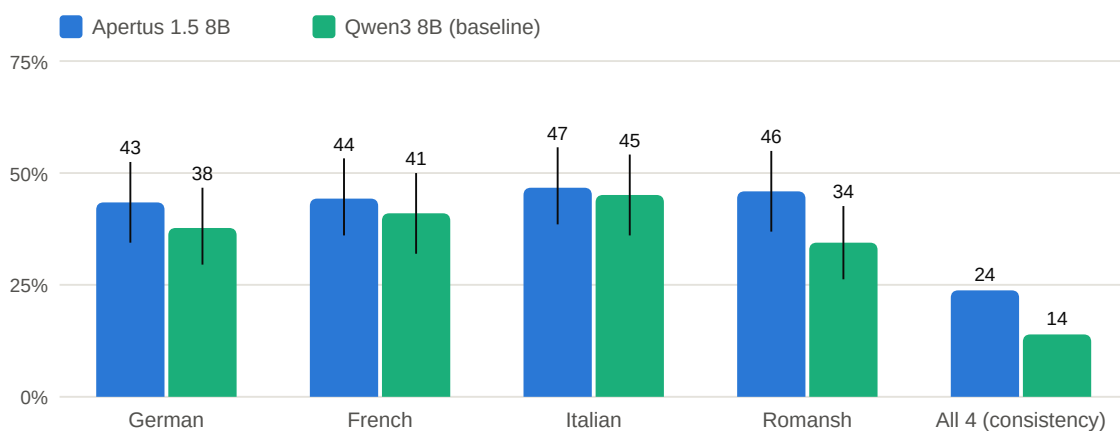


Figure 3: Accuracy per language with 95% bootstrap CIs, and consistency@4, for Apertus 1.5 8B and the Qwen3 8B baseline.

Metric	Apertus 1.5 8B	Qwen3 8B
German accuracy	43.4%	37.7%
French accuracy	44.3%	41%
Italian accuracy	46.7%	45.1%
Romansh accuracy	45.9%	34.4%
Consistency@4	23.8%	13.9%
Hallucination rate	53.2%	58.2%
Not attempted	18	26
False premise (mean)	15%	32.5%

Table 3: Apertus vs. the Qwen3 8B baseline on SwissCivicQA-4L.

Paired tests. Differences are Apertus minus Qwen in percentage points, with a 95% paired bootstrap over facts and an exact McNemar test on the items only one model gets right (no correction for multiple comparisons; borderline values should be read as indicative).

- ▶ **Romansh: Apertus ahead** (significant, but close to the threshold). +11.5 pp (CI +0.8 to +22.1, McNemar $p = 0.0488$; 29 items only Apertus gets right vs. 15 only Qwen). Qwen also abstains much more in Romansh (24 vs. 9 items), and its Romansh answers agree least with its other languages.
- ▶ **Consistency: Apertus ahead.** 29 facts are answered correctly in all four languages by Apertus vs. 17 by Qwen ($p = 0.0227$).
- ▶ **German, French, Italian: no reliable difference** (+5.7, +3.3, +1.6 pp; all CIs include zero). Overall: +5.5 pp (McNemar $p = 0.0364$, fact-level CI -1 to +12.5).
- ▶ **False premises: Qwen clearly ahead.** Apertus rejects the false premise less often (-17.5 pp, CI -30 to -6.3, $p = 0.0043$): Qwen corrects 18 premises Apertus accepts, the reverse happens 4 times.
- ▶ **The agreement signal transfers.** On Qwen, cross-lingual agreement predicts correctness with AUROC 0.76 (Apertus 0.81), so Finding 6 is not specific to one model.

In short, Apertus' Swiss focus pays off exactly where it should, in Romansh and in giving the same answer across the national languages, while its weaker premise checking is a concrete target for alignment.

5. Limitations

- ▶ **Model scope.** Apertus 1.5 8B (text-only Q8_0 on llama.cpp) and one same-size baseline; the 70B model and the original BF16 weights may differ. The harness accepts any OpenAI-compatible endpoint, so this is a one-line rerun.
- ▶ **Judge.** A 12B judge on four languages is imperfect, particularly for Romansh; this is why the grading is audited by rules and by humans and both agreements are reported.
- ▶ **Human audit.** Two volunteer annotators on a 20% sample; Romansh items were judged against the official text and the parallel German item, not by native Romansh speakers.
- ▶ **Translation of questions.** Questions in FR/IT/RM were written with AI assistance; wording quality may vary by language (answers are always grounded in the official text). A Romansh native-speaker review is the most valuable next step.
- ▶ **Contamination.** The Constitution is certainly in Apertus' pre-training data; the benchmark measures recall of in-distribution public knowledge, which is the point for this use case. The question formulations themselves are new.

6. Lessons and recommendations

- ▶ **Report consistency, not only per-language accuracy.** Consistency@4 and pairwise agreement expose gaps that averages hide; we recommend adding them to Apertus' multilingual evaluation.
- ▶ **Train abstention for civic facts.** The hallucination rate on attempted answers is the main risk for public-sector use; preference data that rewards "I am not sure, see Art. X" over a confident wrong number would directly improve this benchmark's F-score.

- ▶ **Cross-lingual alignment data.** The same constitutional provision exists in four parallel official versions. Fine-tuning on aligned question–answer pairs across these versions (the evidence spans in this dataset are a ready starting point) is a cheap way to transfer knowledge the model already shows in one language to the others.
- ▶ **Use the languages as a check.** Querying a second national language and flagging disagreement (Finding 6) catches a large share of wrong answers at inference time, without references or a second model.
- ▶ **Premise checking.** The false-premise subset shows that the model answers the question as asked; a “verify the premise against the Constitution” behaviour is needed before deployment in citizen-facing assistants.
- ▶ **Next iteration of the dataset:** native-speaker review of the Romansh and Italian questions, native-speaker annotators for every language, a stricter judge prompt (the audit shows leniency towards answers with added wrong claims) for partially correct answers, extension to cantonal constitutions and to the 70B model.

7. Reproducibility

`make run` (Docker, < 1 min, no model) re-parses the SHA-256-pinned sources, re-validates every evidence span, rebuilds the dataset, runs the unit tests, recomputes every number in this report from the committed responses, judgments and human labels and checks them against `results/expected/<model>.json`. `make full` downloads both GGUF files (SHA-256 checked), starts pinned llama.cpp containers and regenerates responses and judgments; `MODEL_NAME/MODEL_URL` evaluate any other model. A GitHub Actions workflow runs `make run` on a clean runner for every commit, and the harness is a dependency-free, pip-installable package (`pip install ./track_1b`, Python ≥ 3.10 , CLI `swisscivicqa verify`). Hardware used: 16-core x86 CPU, 32 GB RAM, no GPU.

AI assistance. Fact selection, question drafting, translations and code were produced with substantial help from an AI assistant (Claude, Anthropic); correctness is guaranteed by the mechanical evidence checks and audited by the human review. The model under test and the judge are open-weights models run locally.

Licence. Code Apache-2.0 · dataset CDLA-Permissive-2.0 · report and documentation CC-BY-4.0.

References. Wei et al. (2024), *Measuring short-form factuality in large language models* (SimpleQA), arXiv:2411.04368 · Swiss AI Initiative (2025), *Apertus: Democratizing Open and Compliant LLMs*, arXiv:2509.14233 · Federal Constitution of the Swiss Confederation, SR 101, fedlex.admin.ch, version 2024-03-03.